

01 · ACCESS

Innovators

Students · Startups · Industry
PHONE + OTP

Centre Staff & Managers

Queue · Quotes · Inventory
INSTITUTIONAL SSO

APIS Administration

Utilization across all centres
ROLE-SCOPED

Partner Systems

College ERP · Industry portals
OPENAPI 3.0

02 · CLIENT

Next.js 14 · React · TypeScript · Tailwind

Responsive web (primary) · PWA installable on Android & iOS · Telugu + English
RESPONSIVE WEB · PWA · i18n

SERVER COMPONENTS
STREAMED SEARCH RESULTS

03 · API & SECURITY

REST API · Fastify + Node.js · OpenAPI 3.0 · JWT sessions

TLS 1.3 · RBAC (innovator / staff / manager / admin) · Postgres row-level security · rate limiting · signed URLs for CAD files

04 · CORE SERVICES

FIT — Capability Engine

Answers: which machine can make my part?

- ▶ ROTATION-AWARE BUILD ENVELOPE
- ▶ ACHIEVABLE TOLERANCE FILTER
- ▶ MIN-FEATURE RESOLUTION
- ▶ LIVE MATERIAL STOCK

RANK: AVAILABLE → DISTANCE → FIT

Scheduling

Slot calendar from centre
hours & live bookings

EXCLUDE USING gist

Double-booking rejected
at database level —
not in app code.

No partial slots past close.

Quoting

Versioned, line-itemised:
time · material · setup

source: auto | manual

Staff-priced today.
Every manual quote
trains the engine.
Automates with use.

Job Lifecycle

Explicit state machine

QUEUED → SETUP

→ RUNNING → QC

→ READY → COLLECTED

Append-only event log,
actor + timestamp.
Users watch. Staff move.

Inventory & Ops

Per-machine stock with
reorder thresholds

v_material_alerts

Flags low filament
before a job fails.

Maintenance booked
like any job — logged.

05 · DATA

PostgreSQL 16

centre · machine · machine_capability · material
booking (tstzrange) · job · quote · job_event

CHOSEN FOR tstzrange + EXCLUDE USING gist

Read replicas · row-level security

Redis

Slot-hold cache
Session store

CONTENTION RELIEF

S3 (ap-south-1)

CAD · Gerber · drawings
Signed URLs only

IP-SENSITIVE FILES

Analytics — from the same log

v_machine_utilization · v_material_alerts
Booked hours / machine / week · no-show rates

TRACKING DATA *IS* UTILIZATION DATA